

$$\begin{array}{ccc}
 (X \otimes Y) \otimes Z & \xrightarrow{(\text{coev}_X^{Y \otimes Z} \otimes Y) \otimes Z} & ([Y \otimes Z, X \otimes (Y \otimes Z)] \otimes Y) \otimes Z \\
 \downarrow \alpha_{X,Y,Z} & & \downarrow (\beta_{Y,Z,X \otimes (Y \otimes Z)} \otimes Y) \otimes Z \\
 & & ([Y, [Z, X \otimes (Y \otimes Z)]] \otimes Y) \otimes Z \\
 & & \downarrow \text{ev}_{[Z, X \otimes (Y \otimes Z)]}^Y \otimes Z \\
 X \otimes (Y \otimes Z) & \xleftarrow{\text{ev}_{X \otimes (Y \otimes Z)}^Z} & [Z, X \otimes (Y \otimes Z)] \otimes Z
 \end{array}$$